

Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously ML on graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,444 citations; h-index 15 (August 2026).

Education

Imperial College London <i>Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein</i> Research focused on graph machine learning methods for large real-world systems	<i>Oct 2020 – Feb 2024</i>
University of Cambridge <i>Master in Advanced Computer Science, Distinction (4.0 GPA)</i>	<i>Oct 2018 – Jun 2019</i>
Imperial College London <i>Bachelor in Computer Science, First Class Honors (4.0 GPA)</i>	<i>Oct 2015 – Jun 2018</i>

Experience

Postdoctoral Researcher <i>GLADIA Research Group</i> 🔗 , Sapienza University of Rome – With Prof. E. Rodolà	<i>Rome (remote)</i> <i>Oct 2025 – Present</i>
Developing novel machine-learning methods to deepen our understanding of non-human communication.	
Machine Learning Researcher <i>VantAI</i>	<i>NYC (remote)</i> <i>Jan 2024 – Aug 2025</i>
Co-led development of Neo-1 🔗, a state-of-the-art all-atom latent diffusion model for multimodal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.	
Machine Learning Researcher <i>Twitter</i>	<i>London</i> <i>Jun 2019 – Feb 2023</i>
Developed graph ML models for Twitter’s social graph and deployed a recommendation model that generated 1.2M additional daily follows	
Machine Learning Researcher <i>Fabula AI</i>	<i>London</i> <i>Mar 2019 – Jun 2019</i>
Developed graph ML models for fake news detection as part of Fabula AI’s small research team before Twitter acquired the company 🔗	
Software Engineering Intern <i>Google</i>	<i>California</i> <i>Jul 2017 – Oct 2017</i>

Selected Publications

[Full list](#)

Animal Communication & Audio

- *D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà, and E. Rossi. Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models* [🔗](#). *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.
- *F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, E. Rossi et al. OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. Under Review 2026.*
- *E. Rossi and E. Rodolà. Communicating Sound Through Natural Language* [🔗](#). *arXiv 2026*.

Graph Learning

- *E. Rossi et al. Temporal Graph Networks for Deep Learning on Dynamic Graphs* [🔗](#). *ICML GRL Workshop 2020. 1,700+ citations and 1,000+ GitHub stars* [🔗](#).
- *E. Rossi*, F. Frasca* et al. SIGN: Scalable Inception Graph Neural Networks* [🔗](#). *ICML GRL Workshop 2020. 600+ citations and used in production at Airbnb* [🔗](#).
- *B. Chamberlain, S. Shirobokov, E. Rossi et al. Graph Neural Networks for Link Prediction with Subgraph Sketching* [🔗](#). *ICLR 2023, Oral (top 5%). 200+ citations.*

Leadership

Co-Founder, LeadTheFuture (Forbes article 🔗)	<i>Jun 2018 – Mar 2023</i>
Built a non-profit mentoring community with more than 200 mentors and 500 selected mentees	